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Pore size distributions based on AFM imaging and retention of multidisperse polymer solutes Characterisation of polyethersulfone UF membranes with dopes containing different PVP

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Abstract

Several polyethersulfone ultrafiltration membranes have been made with small quantities of polyvinyl-pyrrolidone of different molecular weights. Our aim was to increase the permeability without significant changes in selectivity.

Different approaches to get effective pore size distributions have been compared between them and with the topographic results obtained from atomic force microscopy (AFM) image analysis. In particular, the use of cumulative retention coefficients have proved to be preferable to adequately obtain effective pore size distributions based on retention of multidisperse solutions.

Within this frame, it has been proved that small PVP content of the studied membranes leads to different increasing permeabilities with very similar selectivities. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Ultrafiltration membranes; Pore size distributions; Retention tests; AFM topography; Membrane manufacture

1. Introduction

Appropriate elucidation of structure is not only relevant to describe sieving effects but also to study solute–material interactions, as far as the corresponding interfaces are placed inside the pores. Thus, electrically determined membrane properties act on the solutes inside the pores, and transport is affected by these properties (zeta potential, surface charges, etc.) in such a way that makes necessary a detailed knowledge of pore geometry to adequately correlate interactions with their effects on flux.

The methods based on solute retention are extraordinarily interesting from the application point of view. This is so because the factors to be considered to translate actual geometrical or structural characteristics into solute-retention performances are especially complex (friction, elasticity, electrical or chemical interaction, hydrophilicity or hydrophobicity, diffusion into the solid matrix, etc.). A first approximation should consider sieve factors as the only or the most relevant phenomena by avoiding to take into account friction or elasticity (mechanically or chemically determined)

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Nomenclature

a	maximum value of Eq. (13) (dimensionless)
A_p	pore area (m)
B	constant of the logical curve (Da)
c_0	feed concentration given as a volume fraction (dimensionless)
c_m	membrane concentration in contact with the high-pressure interface given as a volume fraction (dimensionless)
c_{m_i}	membrane concentration given as a volume fraction in contact with the high-pressure interface for each molecular weight of a polydisperse mixture (dimensionless)
c_p	permeate concentration given as a volume fraction (dimensionless)
c_{p_i}	permeate concentration given as a volume fraction for each molecular weight of a polydisperse mixture (dimensionless)
C	exponent of the logical curve (dimensionless)
d_p	pore diameter (m)
f	frequencies at the lognormal distribution function (dimensionless)
J_s	solute flux (m/s)
J_V	volumetric flux (m/s)
$J_{V,t}$	volumetric flux transmitted through the non-rejecting fraction of pores (m/s)
J_w	pure water flux (m/s)
$J_{w,r}$	pure water flux transmitted through the rejecting fraction of pores (m/s)
$J_{w,t}$	pure water flux transmitted through the non-rejecting fraction of pores (m/s)
K	normalisation constant in Eq. (10) (m^4)
M_w	molecular weight of the solute (Da)
N	cumulative pore density (m^{-2})
N_T	total pore density (m^{-2})
Δp	pressure drop through the membrane (Pa)
R	true retention coefficient (dimensionless)
R_a	average roughness (m)
R_i	true retention coefficient for each molecular weight (dimensionless)
R_i^a	accumulate true retention coefficient for a polydisperse mixture of solutes defined in Eq. (12) (dimensionless)
R_o	observed or apparent retention coefficient (dimensionless)
$\eta(0)$	solvent viscosity (kg/m s)
z_i	z-coordinate in the AFM pictures (m)
z_m	mean value of the z-coordinate in the AFM pictures (m)

Greek symbols

$\eta(c_m)$	solution viscosity at concentration c_m (kg/m s)
μ	mean of the lognormal distribution (units of x)
σ	standard deviation of the lognormal distribution (dimensionless)

appearing in the hydrodynamical interaction of solute molecules and membrane material. In any case, this is equivalent to consider an effective pore size as a result of the action of such factors. In this way, if a more or less polydisperse solution is filtered through a membrane and its molecular weight and/or size distribution are analysed up- and downstream from which an effective pore size distribution can be obtained. On the other hand, the correlation of molecular weight and sizes depends on both the solute–solute and solute–solvent interactions. These, say, colloidal properties make necessary to fix standard conditions of concentration, solvent type, pH, ionic strength, etc.

On the other hand, the process of mass transfer on the membrane is determined by several factors.

1. Adsorption and fouling on the membrane surfaces during the process [1].
2. Formation of a concentration polarisation layer due to solute accumulation on the high-pressure side of the membrane with retro-diffusion and possible gelification [2].
3. Other limiting factors, such as osmotic pressure, may appear too [3].

Among these factors, the most difficult to quantify and thus to be taken into account are both adsorption and fouling. In order to approximate ourselves as much as possible to the actual structural characterisation of the membrane, low-interacting solutions should be used under conditions leading to low-fouling levels.

The concentration polarisation layer can be analysed by using several different models, but none of them gives good results, neither for all membranes (reverse osmosis, nanofiltration, ultrafiltration or microfiltration ones) nor under all hydrodynamic

conditions, on the retentate surface. Nevertheless, the so-called film model gives good results in spite of its simplicity for membranes in the reverse osmosis to ultrafiltration ranges [4]. Some precautions should be followed as well to ensure its applicability. (1) Mass transfer through the membrane system is limited by the concentration polarisation layer more than by the whole membrane. (2) Cross-flow (tangential flow) filtration devices should be used. (3) The retentate should be recirculated onto the membrane at a high speed.

As a consequence of all these factors, the solute should be chosen: (1) to have low concentrations and (2) to be accessible at a broad molecular weight range without significant changes of their physico-chemical properties.

In order to study nano- and ultrafiltration membranes, aqueous solutions of some linear polymers could be chosen, such as polyethylene glycols or dextranes. Polyethylene glycols should be preferred, as their colloidal properties are only slightly sensitive to chemical environment giving low-fouling levels.

If the retention experiments could be performed with totally monodisperse solutes, then a mass balance should allow deducing the operative or effective pore size distribution as shown later on. Strictly speaking, it is unrealistic to assume monodisperse solutes. The proteins and other biological chemicals closely approximate to these ideal conditions, and have been used in this way [5] by dealing adequately with the high fouling then encountered.

In any case, given that different experiments must be performed to use more or less approximately monodisperse solutions, it is desirable to use broad polydisperse solutions in a single experiment. This has been already recognised by Nobrega et al. [6], who used the existing theory [7] to deal with polydisperse solutions. In fact, it seems necessary to modify appropriately the calculation procedure to fit specific characteristics of the polydisperse experiments.

Membranes with different fluxes but similar selectivities will be used here in order to test the differences in the effective pore size distribution obtained when working with polyethylene glycols in multiple almost monodisperse experiments and a single retention experiment with ad-hoc polydisperse solutions. These polydisperse retention experiments will be analysed both using the standard calculations and modifying adequately the definition of retention coefficients.

The membranes to characterise are three different polyethersulfone ultrafiltration membranes with small quantities of polyvinyl-pyrrolidone of different molecular weights giving different permeabilities without significant changes in selectivity. Several researchers [8–12] have reported the effect of PVP additive on the performance of an ultrafiltration membrane. Boom et al. [13,14] have exhaustively studied the important role of PVP in the phase-inversion process. Findings reveal that the phase-separation mechanisms occur by spinodal decomposition either to form the dense skin as well as the microporous sub-layer.

Lafreniere et al. [10] studied the effects of PVP ($M_w = 10{,}000$) concentration on 15% PES membrane behaviour. They observed that with the increasing amount of PVP, the permeate flux increased reaching a maximum at a PVP/PES ratio of 0.5 and then decreased. The pore size distribution using equations developed from surface-pore flow (SFPF) model [15] was determined. Bimodal distributions with values of average radius between 12.4 Å (0% PVP) and 16.8 Å (40% PVP) and from 51.9 Å (0% PVP) to 30 Å (40% PVP) for the first and second distributions, respectively, were found. The higher distribution dispersion was located at the PVP/PES ratio value of 0.5.

When the PVP concentration was varied from 0 to 40% in 20% PES membrane, the maximum permeate flow was reached at a PVP/PES ratio of 1. Pore size values of 11.8–9.4 Å for the first distribution and 43.5–46 Å for the second one were found. The higher dispersion was for a PVP/PES ratio of 1.

Miyano et al. [12] found bimodal distribution with 20% PES to 2–8% PVP membranes, with average pore size values of 52–30 Å for the first and 162–124 Å for the second distribution, being evident that both the pore size and hydraulic permeability diminish with the increase in the PVP concentration. The PES membrane without additive showed an unimodal distribution with mean pore size of 295 Å.

2. Theory

2.1. Surface pore size distributions by AFM

Atomic force microscopy (AFM) [16–18] is a characterisation technique which presents very high possibilities of development and application in the field of

microscopic observation and characterisation of various surfaces. Several operation procedures can be used in AFM: contact mode, non-contact mode and tapping or intermittent contact mode. The tapping mode maintains the high resolutions achieved in the contact mode, but minimises the surface damage, as far as it eliminates the lateral friction forces.

Some parameters can be obtained directly from the AFM pictures [19,20]; for example, the surface pore density or number of pores per surface unit, N_T, and the porosity or porous surface fraction, ε. In order to get the pore size distribution, other parameters can be selected to study their statistical distributions, namely the pore area A_p along with the Feret pore diameter that can be defined as [19,21–23]

$$d_p = 2\sqrt{\frac{A_p}{\pi}} \tag{1}$$

Thus, the equivalent pore diameter is the diameter that a pore of area A_p should have if it had a circular section in the surface. Consequently, the area of each pore is measured immediately from the microphotographs, and the diameter is calculated by using Eq. (1).

Of course, we have also information on height, which permits the study of pore entrances in greater detail. In this case, it is possible to use flat projections of the surface at different heights along with the level profiles at differently randomly chosen lines [24]. The simultaneous use of images and profiles greatly facilitates identification of the entrance of individual pores (however, the diameter deep in the membrane may not be determined directly by surface AFM due to convolution between the tip shape and the pore). In any case, this direct information on height can be used to get a measure of roughness as (among other possible definitions)

$$R_a = \frac{1}{n}\sum_{i=0}^{n}|z_i - z_m| \tag{2}$$

which gives the so-called average roughness, R_a, in terms of the heights, z_i, for the n points in a measured sample and its mean, z_m.

2.2. Apparent and true retention

An observed or apparent retention coefficient for a solute is defined as

$$R_o = 1 - \frac{c_p}{c_0} \tag{3}$$

where c_p and c_0 are the permeate and feed concentration of the solute, respectively. Actually, due to concentration-polarisation, a true retention coefficient can be defined in terms of the concentration directly in contact with the membrane, c_m ($>c_0$), as

$$R = 1 - \frac{c_p}{c_m} \tag{4}$$

There are several methods that can be used to obtain true retention coefficients in ultrafiltration processes. The most used method consists of working under gelation conditions, i.e. when the volumetric permeate flux, J_V, becomes pressure independent. Under such conditions, the concentration on the membrane can be assumed constant, and the film model can be applied to evaluate the mass transfer and thus the true retention.

Nevertheless, in such conditions, the concentration on the membrane active surface is so high that the solute film layer adds a high resistance to this of the membrane itself, giving higher retention than the actual membrane. This is especially inconvenient because the best efficiencies are obtained below concentration polarisation conditions. These polarisation effects decrease when permeate is re-circulated at high speeds and the pressure drop through the membrane is low enough. Thus, the retention coefficient can be evaluated for different low-pressure drops and thus extrapolated to a zero pressure. Under these extrapolated conditions, the membrane concentration should be very similar to the feed one, and thus the so evaluated retention should refer to the membrane itself (true retention) [25].

2.3. Pore size distribution

In order to evaluate the pore size distributions of a partially retaining membrane, it can be assumed that the retention is due to a sieving mechanism. In such a way that, for each molecular weight, there is a fraction of totally retaining pores, while the rest of them allow a free pass of the molecules [7]. Then, we can write the mass balance for each molecular weight as

$$J_V c_p = J_{V,t} c_m \tag{5}$$

where $J_{V,t}$ is the volumetric flux transmitted through the non-rejection fraction of pores. On the other

hand, the ratio of the transmitted volumetric flux and pure water flux passing through the transmitting pores is

$$\frac{J_{V,t}}{J_{w,t}} = \frac{\eta(c_m)}{\eta(0)} \tag{6}$$

where $\eta(c_m)$ and $\eta(0)$ are the solution and solvent viscosities. But for low c_m, this ratio can be approximated by 1 in such a way that Eq. (5) can be rewritten as [6]

$$\begin{cases} J_{w,t} = J_V(1-R) \\ J_{w,r} = J_V R \end{cases} \tag{7}$$

Therefore, $J_{w,t}$ and $J_{w,r}$ (pure water flux passing through the retentive pores) can be evaluated once J_V for each R is known. Then, by using again the mass balance

$$\begin{cases} J_V c_p = J_s \\ J_V - J_s = J_w \end{cases} \tag{8}$$

where c_p has to be expressed as a volume fraction, J_s the solute flow and J_w the total pure water flow that can be obtained by this equation. Thus, $J_{w,t}/J_w$ versus the molecular weight gives the accumulated fraction of flux passing through the non-rejecting pores. The derivative of this function should thus provide the flux-carrying molecules of a given molecular weight. In order to increase definition, an analytical function can be used to interpolate the $J_{w,t}/J_w$ versus the molecular weight M_w. The derivative of this function can easily be calculated point by point. To reproduce well the experimental data, a logical curve, with horizontal asymptotes at $J_{w,t}/J_w = 1$ and 0, seems appropriate as follows:

$$\frac{J_{w,t}}{J_w} = \frac{1}{[1 + (M_w/B)^C]} \tag{9}$$

where B and C are constants to be evaluated by fitting Eq. (9) to experimental results. In this way, $d(J_{w,t}/J_w)/dM_w$ can be obtained. Singh et al. [23] used a lognormal distribution to fit these data; nevertheless, the procedure used by us seems to be more adaptable.

On the other hand, the abscissas can be changed from M_w to pore sizes, d_p, by considering the gyration radii of the polyethylene glycols (PEG) used here as can be obtained from literature [26–29] and fitted to $d_p = (1550 \pm 60) \times 10^{-13}\ M_w^{1/2}$. These resulting pore sizes could be modified by taking into account the number of adsorption layers onto the pore walls and the molecular ability to pass through a pore, from complete rigidity to some degree of flexibility that should allow a size reduction when crossing the membrane [22]. Thus, these assumptions lead to an evaluation of $d(J_{w,t}/J_w)/d(d_p)$.

Finally, the differential flux fraction can be correlated with the pore fraction through the Hagen–Poiseuille equation in such a way that

$$\frac{d(N/N_T)}{d(d_p)} = \frac{d(J_{w,t}/J_w)}{d(d_p)} \frac{K}{d_p^4} \tag{10}$$

K being a normalisation constant.

When the retention experiments are performed with monodisperse solutes, the retention coefficient is evidently well defined by Eqs. (3) and (4). Then several experiments with solutes with different molecular weights should be performed, ensuring that the whole retention spectrum (from 0 to 100%) was covered. It is clearly more convenient to perform a single experiment with an appropriate polydisperse solute or a mixture of somehow polydisperse similar solutes covering at once the entire retention spectrum [6]. Then, the retention coefficient should be redefined for each molecular weight as, for example by

$$R_i = 1 - \frac{c_{p_i}}{c_{m_i}} \tag{11}$$

where sub-index i refers to a single molecular size (a class in the size distribution) in the solute.

Nevertheless, Eqs. (5)–(9) refer to total transmitted and retained fluxes and are based on mass balances. Thus, it seems reasonable to define the retention coefficient as

$$R_i^a = 1 - \frac{\sum_{k=0}^{i} c_{p,k}}{\sum_{k=0}^{i} c_{m,k}} \tag{12}$$

where $\sum_{k=0}^{i} c_{p,k}$ is the solute concentration of the permeate, due to molecules with sizes equal or smaller than the ith one. The concentration $\sum_{k=0}^{i} c_{m,k}$ refers to the solution on the membrane due to sizes below the ith ones.

3. Experimental

3.1. Membranes

Polyethersulfone (Ultrason E 6020P, M_w = 58,000, $T_g = 225°C$) from BASF Aktiengesellschaft, Germany, was used. Two kinds of PVP were used: K30 from Fluka (M_w = 40,000) and K360 from Sigma (M_w = 360,000). The solvents were *N*,*N*-dimethylformamide (DMF) and tetrahydrofurane (THF) from Merck (analytical grade).

Three kinds of membranes were laboratory prepared following the standard method of phase inversion technique: (a) without additive, (b) with 2 wt.% PVPK30, and (c) with 2 wt.% PVPK360. The PES concentration was 17 wt.% in a casting solution formed by DMF with a 1.5 wt.% of THF.

Casting a polymer solution onto a glass plate made flat membranes. After 30 s of solvent evaporation in ambient atmosphere (20°C), the membranes were immersed in water (20°C) for 1 h. Then the membranes were stored in a glycerine/water solution for 12 h and cleaned with 500 ml water bidistilled prior to be used in the experimental tests.

3.2. AFM microscopy

The membranes have been also structurally characterised by AFM. Images were obtained with a Nanoscope IIIA microscope, from Digital Instruments, in the tapping mode. The scanned surface for each image was chosen to have enough definition of the pore borders along with a high number of pores to assure statistical relevance. Images were obtained at room temperatures with common Nanoprobe™ (a registered trademark from Digital Instruments) tapping tips obtained from etched silicon having a high aspect ratio, thus avoiding or minimising convolution of the tip shape by the membrane surface.

The so-obtained three-dimensional images have been conveniently analysed by means of the Jandel® ScanPro software (version 3.00.0030). Each photograph was digitised with a resolution of 1024 × 768 pixels. In all the cases, first of all we assign to each pixel a grey level ranging from 0 (black) to 255 (white).

The image grey spectrum is spanned to get the maximum contrast and definition. Then the images are redefined according to an assigned grey threshold level under which every pixel is assigned to 1 and the rest to 0. The resulting binary picture is filtered to eliminate isolated pixels and to smooth the pore borders, and then all the remaining 1's in the matrix are assumed to belong to a pore. Of course, a correct selection of threshold grey level is fundamental to perform a correct analysis of accurate assigned pores. Customarily, the grey spectrum is analysed and the threshold placed centred in the peak-to-peak valley of the almost bimodal distributions obtained. Unfortunately, sometimes the spectra are so flat that this technique is only of relative help to make a correct threshold election. In any case, eye inspection facilitates the process of selection of several reasonable threshold candidates whose outcomes are conveniently averaged.

3.3. Filtration experiments

All filtration experiments have been done in a Minitan-S cell from Millipore. The retentate is circulated on the membrane by a regulable peristaltic pump Watson-Marlow 704U/R. The detailed experimental device has been described elsewhere [25].

Besides pure water permeability experiments, the solute retention test has been performed on the studied membranes with 1 g/l polyethylene–glycol solutions of increasing molecular weights. We used PEGs of analytical grade from Fluka AG. Their molecular weights for mixtures have been 2, 3, 4, 6, 10, 12, 20 and 35 kDa with the following proportions: 16.5, 4.6, 5.4, 125.4, 14.3, 3.2, 2.7 and 37.9%. These proportions give an almost flat molecular weight distribution for the mixtures. PEGs with molecular weights 4, 6, 8, 10 and 20 kDa have been used for the single PEG experiments.

When single (theoretically monodisperse) PEG has been used, concentrations have been measured by means of a refractive index detector adequately calibrated. An HPLC equipment from Shimazdu, using also a refractive index detector and provided with a Varian GMPWXL column, has been used to obtain the concentration, both for retentate and permeate for each molecular weight slice, for mixed PEG retention experiments.

4. Results and discussion

In Fig. 1, three-dimensional AFM pictures of the three membrane types studied here are presented. In all of them, the z range is 20 nm per division. It can be seen that surface topography is qualitatively quite similar for the three membranes studied. Small differences are apparent nevertheless, as far as grain size is clearly higher for PES17 than for PES17-K30 and PES17-K360. Moreover, according to Eq. (2), their roughnesses are: 1.77 ± 0.11 nm for PES17, 1.96 ± 0.15 nm for PES17-K30 and 2.27 ± 0.36 nm for PES17-K360.

In fact, some more quantitative information can be obtained once images are analysed. In effect, in Figs. 2–4, normalised differential pore size distributions attending to pore number are presented for PES17, PES17-K30 and PES17-K360. All these distributions have been fitted to lognormal distributions given by frequencies

$$f = a\, \mathrm{e}^{(-1/(2\sigma^2))(\ln(x/\mu))^2} \tag{13}$$

where a is the maximum of f, μ the value of x that makes f maximum (most probable x), and σ the standard deviation or width of the distribution.

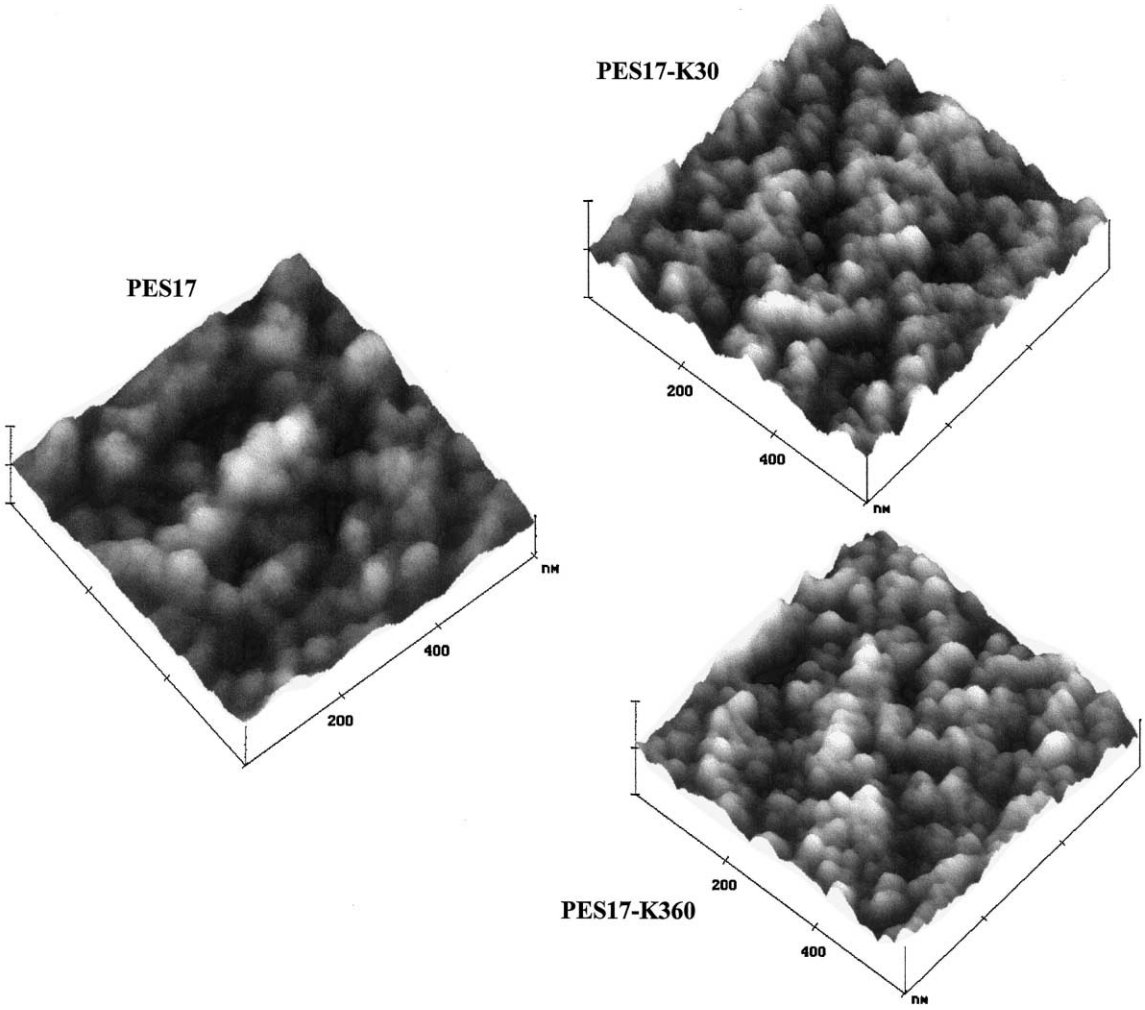


Fig. 1. Three-dimensional AFM pictures of the three membrane types studied here. The z range corresponds to 20 nm per division.

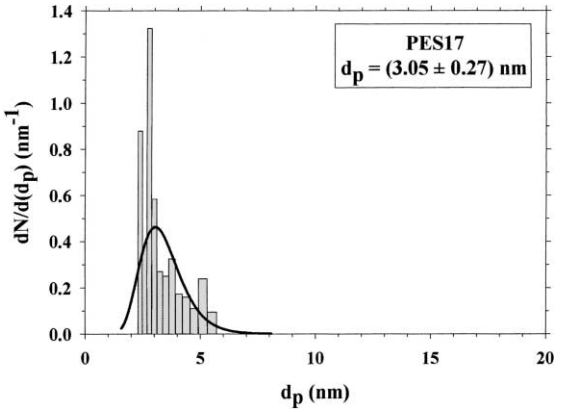


Fig. 2. Normalised differential pore size distribution attending to pore number for PES17 as obtained from image analysis of AFM pictures.

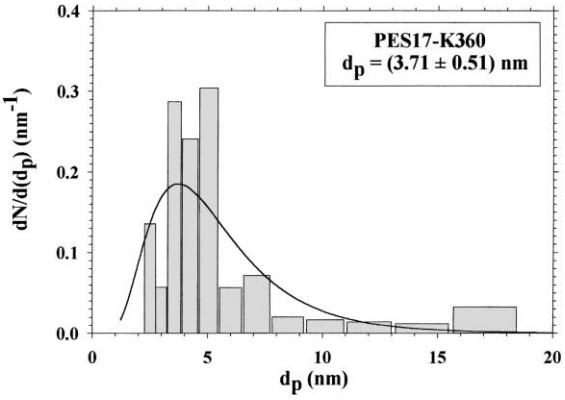


Fig. 4. Normalised differential pore size distribution attending to pore number for PES17-K360 as obtained from image analysis of AFM pictures.

In Fig. 5, R_o is shown versus the applied pressure drop for PES17 along with the corresponding extrapolation done for each molecular weight to obtain the true retention coefficient R. It seems clear that R_o is fairly linear with pressure, thus allowing an accurate extrapolation. On the other hand, the slope is very low which means that we have low concentration polarisation (flux is linear with applied pressure drop), which ensures that R and R_o are very similar and the true retention can be accurately obtained by extrapolation as mentioned before. The results presented here refer to different slices from the PEGS mixture. Quite similar

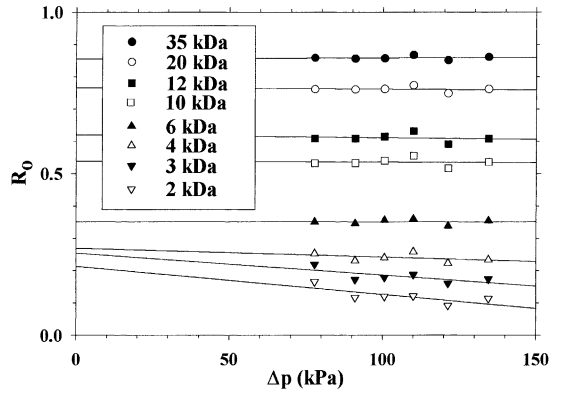


Fig. 5. Observed retention, R_o, vs. the applied pressure drop for PES17. The extrapolation done for each molecular weight is also shown.

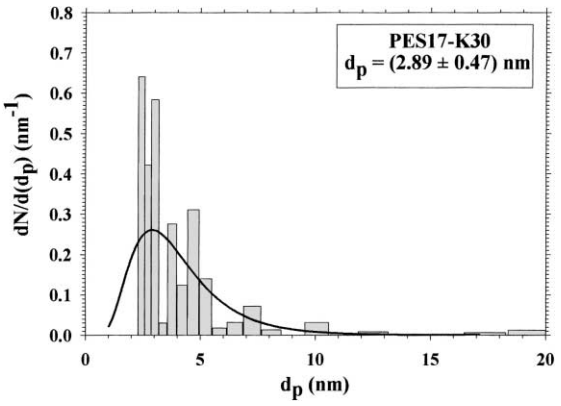


Fig. 3. Normalised differential pore size distribution attending to pore number for PES17-K30 as obtained from image analysis of AFM pictures.

results have been obtained for the other membranes and the relatively monodisperse PEG solutions.

In Fig. 6, the normalised differential pore size distribution, attending to flow, obtained by the different calculation procedures described are shown. It is seen that all the methods give a similar most probable pore size. Both the methods using single monodisperse experiments and non-cumulative retention (Eq. (11)) for polydisperse solutions give almost Gaussian distributions (narrowest for the single monodisperse PEGS). While the method using polydisperse solutions and retentions evaluated according to Eq. (12) gives a clear long-tailed (quite similar to a lognormal) pore size

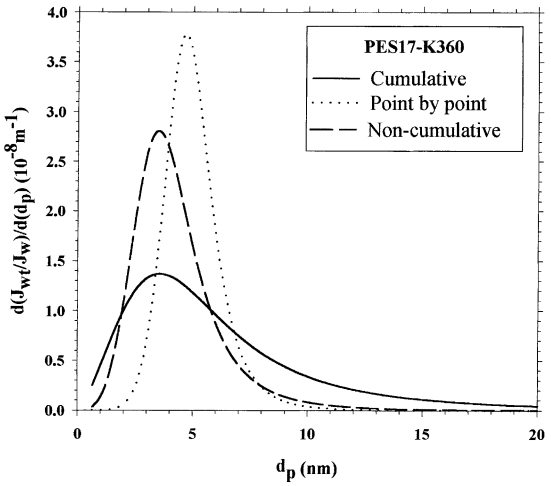


Fig. 6. Normalised differential pore size distribution attending to flow obtained by the different calculation procedures described for PES17-K360.

distribution. All of them have, in fact, been generated according to Eq. (9).

Finally, the normalised differential pore size distribution attending to flow obtained by the Eq. (12) is shown in Fig. 7 for the three membranes studied here. The vertical lines correspond to the most probable pore sizes according to AFM distributions shown in Figs. 2–4.

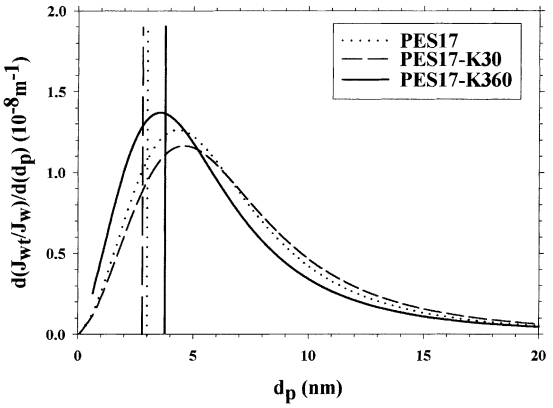


Fig. 7. Normalised differential pore size distribution attending to flow obtained by Eq. (12) for the three membranes studied here. The vertical lines correspond to the most probable pore sizes according to AFM distributions shown in Figs. 2–4. From fittings to lognormals, the following values are obtained: $d_p = 4.06 \pm 0.61$ nm for PES17, 4.32 ± 0.62 nm for PES17-K30 and 3.32 ± 0.67 nm for PES17-K360.

It is clear thus that according to Fig. 7, the three membranes have very similar selectivities in spite of a slight increase in the small-pores fraction for long-chain PVP. On the other hand, the hydraulic permeabilities of the analysed membranes are 2.17×10^{-10}, 5.62×10^{-10} and 7.47×10^{-10} m/s Pa for, respectively, PES17, PES17-K30 and PES17-K360. Given their very similar pore size distributions obtained from the retention experiments, it is clear that the three membranes should have quite different porosity, tortuosity and/or thickness. In particular, according to the Hagen–Poiseuille relation, the ratio $\Theta/\phi\,\Delta x$ (porosity divided by tortuosity and thickness) should be for PES17-K30 2.6 times that of PES17 and 3.4 times for PES17-K360.

It is seen that the AFM pore size distributions are not so similar for the three membranes studied as the retention ones. In effect, they show increasing fractions of big pores as the molecular weight of the PVP added increases. Therefore, it seems clear that the addition of PVP changes mainly the surface porous structure, along with only some bulk parameters as porosity, thickness and/or tortuosity. These changes give increasing permeabilities without changing the membrane selectivity, as intended when changing the manufacturing process. On the other hand, it should be noted that roughness increases, as mentioned, when PVP is added and more for longer PVP chains. Some of the differences in surface pore size distributions (AFM) should appear, because some of the smallest valleys (more for increasing roughness) are inadequately counted as relatively wide pores (see Figs. 2–4).

As far as the reasons for the detected increase in flow by using increasing molecular weights of the added PVP, several more or less speculative reasons can and have been proposed in literature. The presence of the N–C=O group in PVP could increase hydrophilicity of the membrane, thus making usable smaller pores and increasing permeability [10]. On the other hand, the interaction between this PVP group and the O=S=O from PES could cause an increase in the big-pores fraction leading also to higher permeabilities [10]. Other possibility is the appearance of a thinner dense layer with increasing fractions of small pores and macrovoids appearing below the dense layer,

both leading to higher permeabilities. None of these possibilities can be rejected and in fact could act simultaneously to different extents to justify the results obtained here.

5. Conclusions

It has been proved that the addition of small quantities of polyvinyl-pyrrolidone of different molecular weights to a phase inversion ultrafiltration membrane of polyethersulfone resulted in an increase of permeability without significant changes in selectivity.

Different approaches to get effective pore size distributions from retention experiments have been compared. In particular, the use of cumulative retention coefficients have proved to be preferable to adequately obtain effective pore size distributions based on retention of multidisperse solutions. The main advantage of this procedure is that it allows getting complete distributions in a single run. The so-obtained pore size distributions are more accurate than those obtained from single discrete experiments, using theoretically monodisperse solutions. This is due to probably two main factors: the solutions are never monodisperse and only few experiments are performed to simplify the procedure. When using appropriate multidisperse solutions, we take into account and actually we take advantage of polydispersity to have as many points as necessary to get accurate pore size distributions.

AFM seems very adequate to get the most probable pore size, while the complete pore size distribution could be affected by insufficient discrimination between certain valleys and actual pores, mainly when high roughness is present.

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